INCH-POUND

A-A-60063
July-12,-2024

COMMERCIAL ITEM DESCRIPTION

LATHE, ENGINE (STANDARD) AND TOOLROOM (PRECISION), HORIZONTAL SPINDLE

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers horizontal spindle, engine (standard) and toolroom (precision) lathes intended for turning, boring, drilling, reaming, tapping, and facing of workpieces made from cast iron, steel, titanium, aluminum, brass, graphite epoxy, and composite materials for use in Aircraft Metals Technology Shops across the Air Force. The equipment identified herein should be manufactured in the United States of America.
2. CLASSIFICATION. The lathes shall be of the following types, swings, and sizes. The type, swing, and size to be furnished shall be as specified in Table I.

Table I. Lathe types.

Lathe Type	Description	Size (Center-to-Center)	Swing Over Bed
1	Engine (standard)	38-50 in.	14-16 in.
2	Toolroom (precision)	17-18in.	11 in.

Note 1: Lathe sizes, powers, and capacities shall not be less than, nor exceed 10 percent greater than, the requirements stated in Table I.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data that may improve this document should be sent to: AFLCMC/EZSS, 2145 Monahan Way, Area B, Bldg 28 Wright-Patterson AFB, OH 45433 or email engineering.standards@us.af.mil. Since contact information can change, you should verify the currency of this information using the ASSIST Online database at https://assist.dla.mil.

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3. SALIENT CHARACTERISTICS.

3.1 General requirements. The lathe shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. The lathe shall be constructed from materials with sufficient mass and strength and configured in a shape and size that shall support the workpiece and machine components while minimizing element deflections and initiated and/or transmitted vibration during operation. The rotation of the axis of the workpiece shall be horizontal. The longitudinal axes of machine ways supporting the carriage, tailstock, steady rests, or tool rests shall lie in a horizontal plane and shall be parallel to the rotational axis of the workpiece. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair. Access doors or removable panels shall be provided for servicing all components located within the lathe.

3.2 Components. The lathe shall include, as a minimum, the following components:

3.2.1 Base. The base shall support the lathe bed and all components mounted thereon. The base shall provide a means for leveling and anchoring the machine. A swarf-collection area shall be provided to collect and retain turning chips and coolant fluid either adjacent or within the base.

3.2.2 Base anchoring. The lathe shall be delivered with fasteners, adapters, clamps, or other hardware required to affix the base to the floor. Maximum diameter for each hole required in the shop floor shall be 1 inch.

3.2.3 Bed. The bed shall support the machine ways upon with the carriage, tailstock, and tool or workpiece rests travel. The bed shall be configured to allow the chip swarf to either pass through openings in the bed or be diverted to the collection area.

3.2.4 Spindle and spindle drive. The spindle shall be the manufacturer's standard capable of multiple-step or infinitely variable speeds through a belt or gear drive, or a combination of both. The spindle nose shall conform to American Society of Mechanical Engineers (ASME) B5.9. The spindles shall be compatible with provided chucks (see section 3.2.5). The spindle shall conform to the requirements detailed in Table II.

Table II: Spindle requirements.

Lathe Type	Spindle Motor Rating	Spindle RPM Min.	Spindle RPM Max.	Spindle Bore
1	5-10 hp	0-40 rpm	1800-2500 rpm	1.57-3.18 in.
2	3-5 hp	0-50 rpm	4000 rpm	1.25-1.50 in.

Note 2: Spindle ratings, sizes, and speeds shall not be less than, nor exceed 10 percent greater than, the requirements detailed in Table II.

3.2.5 Chucks and collets. The lathe shall be provided with chucks and/or collets as detailed in Table III. Chucks shall feature serrated jaws and shall be of the manual, wrench operated variety. Chucks and collets shall be capable of imparting clamping force necessary to hold the workpiece during turning operations. Adapters and hardware necessary to mount the chucks and/or collets shall be included. Chucks and collets shall conform to American Society of Mechanical Engineers (ASME) B5.9.

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Table III: Chuck and collet requirements.

Lathe Type	3-Jaw Chuck Size	4-Jaw Chuck Size	Collet
1	8-10 in.	10-12 in.	-
2	6 in.	-	5-C (minimum 7-piece set 1/8 in. - 7/8 in. round) by 1/8 in. increments

3.2.6 Machine ways. The machine ways shall be the standard design of the manufacturer. A means shall be provided to prevent coolant and chip swarf from accumulating on machine ways and becoming lodged between sliding parts. Machine ways shall be machined/ground smooth to allow continuous, uninterrupted movement of carriages, guides, and slides for machining operations. Machine ways shall not corrode or degrade under normal operations and maintenance conditions and/or storage for no less than fifteen (15) years from the date of procurement.

3.2.7 Carriage. Carriage (s) shall be mounted on the bed machine ways and shall move parallel to the rotational axis of the workpiece. Carriages shall support movement and use of the cross-slide(s). Carriages shall be a configuration allowing the movement of the cross-slide or tool turret assembly to be on a plane parallel to the spindle axis. Carriages shall move along the workpiece axis on the ways by hand crank or power feed, if equipped.

3.2.8 Cross-slide and tool turret assembly. The cross-slide shall provide tool movement along the axis perpendicular to the workpiece axis and through the swing of the lathe by hand crank or power feed, if equipped. The cross-slide shall be equipped with a graduated compound rest that shall allow 360-degree adjustment of the tool in the tool post with respect to the workpiece axis.

3.2.9 Tool turret. The toolroom lathe shall be provided with a multiple tool holding turret that can be installed on the ways in place of the tailstock. The turret shall have a minimum of four tool positions and shall be indexable with independently adjustable stops for each turret position. The turret shall have a means to feed and retract the indexed tool into the workpiece.

3.2.10 Threading control. Engine (standard) lathes shall have a threading gearbox to complete threading operations. Toolroom (precision) lathes shall have a threading gearbox or digital thread control. The gearbox power-feed leadscrew shall be capable of being coupled to the spindle to coordinate feed and rotation for cutting threads. The gearbox mechanism shall be provided with safety overload protection. Digital threading control shall utilize a computer-controlled, servo motor leadscrew to complete threading operations. At a minimum, the threading controls shall provide a selectable range of threads within the ranges detailed in Table IV.

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Table IV: Threading control requirements.

	Engine (standard)	Toolroom (precision) with Gearbox	Toolroom (precision) with Digital Control
Inch Thread Range	2-72 TPI	11-108 TPI	3.175-500 TPI
Inch Thread Pitch Minimum Quantity	36	36	-
Metric Thread Range	0.2-14 mm	0.275-2.7 mm	0.05-6.55 mm
Metric Thread Pitch Minimum Quantity	17	36	-

3.2.11 Tailstock. The lathe shall have a tailstock that is adjustable and lockable along the ways. The tailstock spindle travel shall be controlled by a handwheel, have a quick-locking feature, and have a graduated scale to indicate spindle location. The tailstock spindle shall have a tool-holding Morse taper. The tailstock spindle shall have a built-in quick release method for tools.

3.2.12 Safety devices. An emergency stop switch/button shall be provided that when pressed will instantly stop all motion of the machine, servomotors, tool changers, and coolant pump. Torque-limiting capabilities shall be included, at a minimum, on the carriage assembly and cross slide. Engine (standard) lathes shall have an interlocked chuck guard to prevent access to the chuck when in locked operation.

3.2.13 Manual controls. The lathe manual controls shall include, as a minimum, the following functions:

a. Emergency stop.

b. Power on/off.

c. Spindle start/stop.

d. Spindle-speed.

e. Carriage motion.

f. Cross-slide motion.

g. Tailstock motion.

3.2.14 Digital readout system. The lathe shall be equipped with a digital readout system consisting of sensors and display for the X- and Z-axes of travel. The digital readout system shall have a resolution of not greater than 0.0001 inches.

3.2.15 Lubrication system. Machine components requiring continuous lubrication shall be permanently lubricated or lubricated by automatic means. Machine components requiring periodic manual lubrication shall be readily accessible.

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3.2.16 Coolant system. The lathe shall be equipped with a tool coolant system. The coolant system shall consist of a reservoir, motor-driven pump, and chip tray. The system shall have means for draining and cleaning and shall permit the operator to direct and control the amount of coolant flow.

3.2.17 Electrical system. The electrical system shall conform to the National Fire Protection Association (NFPA) 79 standard. The system shall allow for either 230 VAC three phase or 460 VAC three phase connections.

3.2.18 Electric motors. The spindle-drive motor and motion-control motors, if equipped, shall have sealed and permanently lubricated ball or roller bearings, shall be energy efficient, and shall conform to the National Electrical Manufacturers Association (NEMA) MG 1 and NEMA ICS 16 standards, respectively. The spindle-drive motor shall have an open drip-proof enclosure. Motion- and position-control motors and controls, if equipped, shall conform to NEMA ICS 1 and NEMA ICS 3.

3.2.19 Tooling, attachments, and accessories. All tooling, attachments, accessories, and hardware required to render the machine usable shall be included, including, as a minimum:

3.2.19.1 Toolholders. A minimum of two removeable toolholders shall be provided. Toolholders shall be functional with the lathe or toolholder provided. The toolholder shall be capable of holding C-style inserts via top clamping and hole clamping, though not simultaneously. The toolholder shall be in a right-handed, external tool orientation and have a 95-degree angle. The toolholder shall conform to ISO 13399 standards.

3.2.19.2 Steady rest. A removeable steady rest shall be provided for increased support when turning long or narrow workpieces. The steady rest shall be able to be positioned anywhere along the work bed in the Z direction and mechanically locked into place. The steady rest shall be usable with or without the tailstock in use. The steady rest shall be adjustable to support workpieces up to 3.0 inches (toolroom lathes) or 6.375 inches (engine lathes).

3.2.19.3 Quick-change tool kit. A removeable quick-change tool post shall be provided that allows for quick and repeatable manual tool changes. The kit shall include all hardware for mounting the tool post onto the system, for at least four turn/face toolholders compatible with the toolholder and a boring bar holder compatible with the boring bar. The tool post shall be capable of rotating/adjusting orientation and locking into place.

3.3 Warranty. A minimum one-year warranty shall be provided to protect against defects, breaks, or component failures during normal operations. The warranty shall include the replacement and installation of components/assemblies at the installation location by an OEM-certified technician. The warranty does not cover tooling identified in 3.2.19.1 through 3.2.19.4. For equipment located in austere operating environments, an OEM-certified technician may oversee the repair by a Government representative; however, replacement components shall still be provided as part of the warranty.

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3.4 Support. Technical support for the equipment shall be provided for a period of three years after the date of purchase. Technical support shall be accessible, at a minimum, via phone and email.

3.5 Performance. The lathe shall meet the performance requirements of sections 5.3.1 through 5.3.3.

3.6 Safety and health requirements. The machine shall be designed and manufactured in accordance with ANSI B11.6. The manufacturer shall ensure that the machine and all equipment and accessories used on the machine shall comply with Occupational Safety and Health Administration (OSHA) 29 CFR PART 1910. If a conflict arises between the ANSI B11.6 and OSHA standards, the OSHA standards shall apply.

3.7 System of units. The U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. Measuring and indicating devices shall be graduated in both the US and SI system of measurements.

3.8 Nameplate. A nameplate shall be securely attached to the machine and shall contain the following information:

a. Nomenclature.

b. Manufacturer's name.

c. Serial number.

d. Machine model designation.

e. Power input (volts, total amperes, phase, and frequency).

f. Short-circuit/over-current rating.

g. Contract number or order number.

h. National stock number.

i. Date of manufacture.

3.9 Lubrication plate or chart. A lubrication plate or chart shall be attached to the machine. The information provided on the plate or chart shall include:

a. Points of lubricant application.

b. Servicing interval.

c. Type of lubricant(s) with SAE number or lubricant identifier.

4. REGULATORY REQUIREMENTS.

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

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4.2 Environmental protection. The item shall meet all applicable Environmental Protection Agency (EPA) restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user's health and shall be adhered to during the manufacturing, service, transportation, storage, and operation/use of the item.

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The products shall meet the salient characteristics (section 3) of this commercial item description; conform to the manufacturer's own drawings, specifications, standards, and quality assurance practices; and be the same system offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

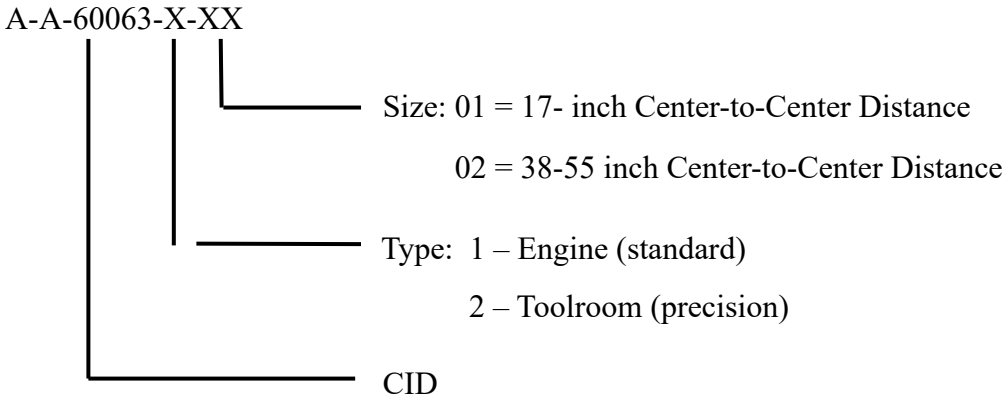
5.2 Acceptance. Upon delivery, the customer shall conduct a thorough inspection to ensure no damages and/or all required tooling/accessories are included with the equipment. The customer shall contact the vendor if any significant damages are discovered, or if there are any missing tooling and/or accessories.

6. PACKAGING.

6.1 Preservation, packing, and marking. The products shall be preserved, packed, and marked as specified in the contract or purchase order.

7. NOTES.

7.1 Part identification number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.



7.2 Sources of referenced documents.

7.2.1 The American Society of Mechanical Engineers (ASME). Copies of ASME documents may be obtained at https://www.asme.org/.

7.2.2 American National Standards Institute (ANSI). Copies of ANSI documents may be obtained at https://www.ansi.org/.

7.2.3 Environmental Protection Agency (EPA). Copies of EPA documents may be obtained at https://www.epa.gov/.

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7.2.4 National Electrical Manufacturers Association (NEMA). Copies of NEMA standards may be obtained at https://www.nema.org/.

7.2.5 National Fire Protection Association (NFPA). NFPA documents may be obtained at https://www.nfpa.org/.

7.2.6 International Organization for Standardization (ISO). Copies of ISO Standards may be obtained at https://www.iso.org/.

7.2.7 Occupational Safety and Health Administration (OSHA). Copies of OSHA documents may be obtained at https://www.osha.gov.

7.2.8 Federal Acquisition Regulation (FAR). Copies of the FAR may be obtained at https://www.acquisition.gov/.

7.3 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this document.

b. Type and size of lathe required (see section 2).

c. Preservation, packing, and marking requirements.

7.4 Key words.
Boring
Drilling
Reaming
Turning

MILITARY INTERESTS:

Custodians:
Air force – 184

Preparing Activity:
Air Force – 184

Reviewers:
GSA – FAS

Agent:
AF – 110

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